

João Pedro Valente de Oliveira Coelho | Curriculum Vitae

DPhil in Evolutionary Anthropology, University of Oxford

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Current research interests: paleoecology, machine learning applications to biological anthropology, hominin origins, geospatial palaeontology, geometric morphometrics, evolution of bipedalism

Professional interests: mentoring and training, science content for the public, virtual learning experiences, 3D-modelling and 3D-printing, digitalization of museum collections

Education and Training

- 2017-2024** **DPhil in Evolutionary Anthropology** – University of Oxford (United Kingdom)
Project: From satellite pixels to fossil apes: targeting gaps in the hominid fossil record
Supervisors: Prof. Susana Carvalho (University of Oxford) and Prof. Robert L. Anemone (The University of North Carolina at Greensboro)
- 2013-2015** **Master's Degree in Human Evolution and Biology** – Universidade de Coimbra (Portugal)
Dissertation: Unwarping Heated Bones: a quantitative analysis of heat-induced skeletal deformations using 3D geometric morphometrics
Supervisors: Dr. David Gonçalves and Prof. Eugénia Cunha (Universidade de Coimbra)
- 2012-2013** **Erasmus programme** – Háskóli Íslands (Iceland)
- 2010-2013** **Graduation in Anthropology** – Universidade de Coimbra (Portugal)

Work Experience

- March 2024 – Current** Paleontology Manager | **Science Department, Gorongosa National Park**, Mozambique
- Lead fieldwork and conduct research for paleontology and primatology projects.
 - Identify funding opportunities and develop grant proposals to support research initiatives.
 - Manage collections and train staff to optimize the Paleontology Lab's operations.
- 2018-2024** Field School Instructor | **PaleoPrimate Project Gorongosa**, Mozambique
- (during summer breaks)
- Experience with laboratory and database management, photography, 3D photogrammetry, fossil curation, anatomy and taxonomy.
 - Co-directed a paleontological excavation (GPL-12-E in 2022, GPL-12-W in 2023).
 - Primatology (bipedalism and carnivory) and Taphonomy (carcasses monitoring) data collection for the 'Bones of Predation' project (2021).
 - Surveying team leader (in 2018 and 2019).
- 2017-2024** Graduate Student and DPhil Candidate | **PrimoBevoLab, Institute of Human Sciences**, Oxford
- Researcher at the *Primate Models for Behavioural Evolution Lab*, University of Oxford.
 - Co-organized an international conference, and created its website - EFP/PSGB2019.
 - Data collection at Nairobi National Museum, Kenya and Oxford's Natural History Museum.
 - Led the *Miocene Apes and the Origins of Bipedalism* Seminar for the MSc Cognitive and Evolutionary Anthropology (cohorts of 2022 and 2023).
 - Tutorial's instructor (on genetics and molecular clocks).
 - Teaching (undergrad students in archaeology and anthropology BSc from Oxford).
- 2016-2017** Co-founder, webmaster and R instructor | **JEST - Junior Enterprise for Science and Tech**, PT
- data modelling and visualization for life and health sciences
 - I co-founded JEST - Junior Enterprise for Science and Tech @ University of Coimbra.
 - Head designer and webmaster of the original project (jest.pt).
 - Organized workshops and conferences on data science, and programming (R and Python).
 - Instructor for several R workshops (intro, data viz, modelling, AI, ML, shiny, etc.).
- 2015-** Researcher | **Centre for Functional Ecology**, Coimbra (Portugal)
- Collaborator at the 'Forensic Anthropology and Paleobiology' group – cfe.uc.pt.

- 3D scanning expert and curator of the CEIXXI Digital Museum (see sketchfab.com/lfa).
- Webmaster and Web Designer at 'Laboratory of Forensic Anthropology' – lfa.uc.pt.
- Statistical shape analysis (geometric morphometrics) of osteological material.
- Public communication and workshops, science communication.

2013-

Founder, software developer, designer | **Osteomics** (startup venture)

I develop and maintain Osteomics—a web platform offering free software tools for bioanthropologists. We provide consulting on statistical analysis and app development for scientific projects in forensic anthropology and human evolution.

- Lead Designer
- Web App Developer
- R and Shiny statistical products
- 3D Printing and 3D Scanning services

Fieldwork Experience (39 weeks total)

2024 (6 weeks)	Gorongosa, Mozambique – Associate Director for the Gorongosa Interdisciplinary Field School in Human Evolution, and Organizer of the November Geological Expedition.
2023 (4 weeks)	Gorongosa, Mozambique – Instructor at the Oxford-Gorongosa Paleo-Primate Field School (University of Oxford) + Excavation co-direction + Lab work (taxonomy).
2022 (6 weeks)	Gorongosa, Mozambique – Co-directed a paleontological excavation + Instructor of the Evolution module at the Master's in Conservation Biology + Lab work (taxonomy).
2021 (9 weeks)	Gorongosa, Mozambique – Collected data on a primatology (bipedalism) and a taphonomy (predation to fossilization) project as a member of PaleoPrimate Project Gorongosa.
2019 (4 weeks)	Gorongosa, Mozambique – Instructor at the Oxford-Gorongosa Paleo-Primate Field School (University of Oxford); PhD research at Gorongosa National Park.
2019 (6 weeks)	East Turkana, Kenya – Graduate Assistant at the Koobi Fora Field School (George Washington University); member of the Paleoecology research group at Koobi Fora.
2018 (4 weeks)	Gorongosa, Mozambique – Instructor at the Oxford-Gorongosa Paleo-Primate Field School (University of Oxford); PhD research at Gorongosa National Park.

Honours and Awards

2020/21	The Boise Trust Fund Research Award – Department of Zoology, University of Oxford
2020	Best Poster Prize - Primate Society of Great Britain - Winter meeting 2020
2018	The Boise Trust Fund Research Award – Department of Zoology, University of Oxford
2017	The Boise Trust Fund Research Award – Department of Zoology, University of Oxford
2017-2021	PhD Scholarship SFRH/BD/122306/2016 – <i>Fundação para a Ciência e a Tecnologia</i> , Portugal
2014	Finalist: "Best Business Plan" for Osteomics – <i>Arrisca C</i> 2014, Universidade de Coimbra
2013-2014	Grant recipient for a biotech startup – <i>Passaporte para o Empreendedorismo</i> , IAPMEI
2013	Finalist: "Best Business Idea" for Osteomics – <i>Arrisca C</i> 2013, Universidade de Coimbra
2012-2013	Scholarship: LLP/Erasmus Programme – <i>Háskóli Íslands</i> (University of Iceland)

Publications

- Carvalho S, Spagnoletti N, **d'Oliveira Coelho J**, Beardmore-Herd M, Mendonça-Furtado O, Verderane M, Prieto J, Cavallaro DI, Izar P, Visalberghi E. 2026. Small vertebrate consumption by capuchin monkeys (*Sapajus libidinosus*): Insights into the human predatory pattern. *PLOS One* 21:e0355204.
- Carvalho S, Bobe R, Beardmore-Herd M, **d'Oliveira Coelho J**, Pascual-Garrido A, Kingdon J. 2026. Natural history is more relevant than ever for primate ecology and conservation. *Primates*. 10.1007/s10329-026-01287-8
- d'Oliveira Coelho J**, Anemone RL, Bobe R, Carvalho S. 2026 Parting ways: Pan-Homo divergence revisited. *Primates*. DOI: 10.1007/s10329-026-01269-w
- Allen G, Freymann E, **d'Oliveira Coelho J**, Shagara H, Shinyo I, Panda A, Jaya A, Hockings KJ, Morrogh-Bernard HC. 2026. Investigating medicinal resource combinations in the Bornean orangutan diet. *Scientific Reports* 16:18690. DOI: 10.1038/s41598-026-52614-4
- Carvalho S, Anemone RL, **d'Oliveira Coelho J**, Bobe R. 2025. The primates of Gorongosa National Park, Mozambique. *American Journal of Biological Anthropology*, **188**(2):e70143. 10.1002/ajpa.70143

- Farassi R, **d'Oliveira Coelho J**, Carvalho S. 2025. Habitat use and the demographics of object manipulation by wild chacma baboons. *American Journal of Biological Anthropology*, 187:e70094. DOI: 10.1002/ajpa.70094
- Biro D, Muschinski J, Hammond P, Bobe R, Bamford M, Capelli C, **d'Oliveira Coelho J**, Farassi R, Lüdecke T, Martinez F, Mathe J, Ferreira da Silva MJ, Carvalho S. 2025. West Side Story: Regional inter-troop variation in baboon bark-stripping at Gorongosa National Park, Mozambique. *American Journal of Biological Anthropology*, 187:e70057. DOI: 10.1002/ajpa.70057
- Regala FT, Pinto MJ, Paulo LM, Bobe R, Raja M, Aldeias V, Archer W, Braun DR, Lüdecke T, **d'Oliveira Coelho J**, Mathe J, Macamo S, Bamford M, Martinez FI, Püschel TA, Capelli C, da Silva MJF, Anemone RL, Carvalho S. 2024. Caves of Inhanga: the speleological heritage of the Cheringoma Plateau, Mozambique. *Cave and Karst Science*, 51:67–80.
- Scott GR, Navega DS, **d'Oliveira Coelho J**, Vlemincq-Mendieta T, Kenessey D, Irish JD. 2024. rASUDAS2: A new iteration of the application for assessing the population affinity of individuals by tooth crown and root morphology. *Forensic Anthropology*, 8(1):1–9. DOI: 10.5744/fa.2024.0006
- Freyman E, **d'Oliveira Coelho J**, Muhumuza G, Hobaiter C, Huffman MA, Zuberbühler K, Carvalho S. 2024. Applying Collocation and APRIORI Analyses to Chimpanzee Diets: methods for investigating non-random food combinations in primate self-medication. *American Journal of Primatology*. e23603 10.1002/ajp.23603
- Bobe R, Aldeias V, Alemseged Z, Anemone RL, Archer W, Aumaitre G, Bamford MK, Biro D, Bourlès DL, Boyd MD, Braun DR, Capelli C, **d'Oliveira Coelho J**, Habermann JM, Head JJ, Keddadouche K, Kupczik K, Lebatard A-E, Lüdecke T, Macôa A, Martínez FI, Mathe J, Mendes C, Paulo LM, Pinto M, Presnyakova D, Püschel TA, Regala FT, Sier M, da Silva MJF, Stalmans M, Carvalho S. 2023. The first Miocene fossils from coastal woodlands in the southern East African Rift. *iScience*. 26 (9). DOI: 10.1016/j.isci.2023.107644
- Curate F, Navega D, Cunha E, **d'Oliveira Coelho J**. 2022. DXAGE 2.0 — adult age at death estimation using bone loss in the proximal femur and the second metacarpal. *International Journal of Legal Medicine*. DOI: 10.1007/s00414-022-02840-y
- Bobe R, **d'Oliveira Coelho J**, Carvalho S, Leakey M. 2022. Early Hominins and Paleoecology of the Koobi Fora Formation, Lake Turkana Basin, Kenya. In: Reynolds, SC, Bobe, R. (Eds.), *African Paleoecology and Human Evolution*. Cambridge University Press, Cambridge: 311–331. DOI: 10.1017/9781139696470.026
- d'Oliveira Coelho J**, Anemone RL, Carvalho S. 2021. Unsupervised learning of satellite images for enhancing fossil site discovery at Gorongosa, Mozambique. *PeerJ* 9:e11573 DOI: 10.7717/peerj.11573
- Curate F, **d'Oliveira Coelho J**, Silva AM. 2021. CalcTalus: an online decision support system for the estimation of sex with the calcaneus and talus. *Archaeological and Anthropological Sciences*. 13:74.
- d'Oliveira Coelho J**, Curate F, Navega D. 2020. Chapter 5.1 - Osteomics: Decision support systems for forensic anthropologists. In: Obertová, Z., Stewart, A., Cattaneo, C. (Eds.), *Statistics and Probability in Forensic Anthropology*. Academic Press: 259–273.
- Rodrigues H, Ramos R, Fagundes L, Galego O, Navega D, **d'Oliveira Coelho J**, Alves FC, Cunha E. 2020. Mastoid, middle ear and inner ear analysis in CT scan – a possible contribution for the identification of remains. *Medicine, Science and the Law*, 60:102–111.
- d'Oliveira Coelho J**, Curate F. 2019. CADOES: An interactive machine-learning approach for sex estimation with the pelvis. *Forensic Science International*, 302:109873.
- Ammer S, **d'Oliveira Coelho J**, Cunha E. 2019. Outline Shape Analysis on the Trochlear Constriction and Olecranon Fossa of the Humerus: Insights for Sex Estimation and a New Computational Tool. *Journal of Forensic Sciences*, 64(6):1788–1795.
- Vilas-Boas D, Wasterlain SN, **d'Oliveira Coelho J**, Navega D, Gonçalves D. 2019. SPINNE: An app for human vertebral height estimation based on artificial neural networks. *Forensic Science International*, 298:121–130.
- Ferreira MT, Coelho C, **d'Oliveira Coelho J**, Navega D, Wasterlain SN. 2018. New data about sex and age-at-death based on the postcranial skeleton of the enslaved adult Africans found at Lagos, Portugal (15th-17th centuries). *Cadernos do GEEVH*, 7(1):7–16.
- Silva A, Tomé T, Cunha C, **d'Oliveira Coelho J**, Valera A, Filipe V, Scott GR. 2018. Unilateral absence of mandibular condyle in a Bronze Age male skeleton from Portugal. *International Journal of Paleopathology*, 22: 168–172.
- Scott GR, Pilloud M, Navega D, **d'Oliveira Coelho J**, Cunha E, Irish J. 2018. rASUDAS: A New Web-Based Application for Estimating Ancestry from Tooth Morphology. *Forensic Anthropology*, 1:18–31.
- Gonçalves D, **d'Oliveira Coelho J**, Amarante A, Makhoul C, Oliveira-Santos I, Navega D, Cunha E. 2018. Dead weight: Validation of mass regression equations on experimentally burned skeletal remains to assess skeleton completeness. *Science & Justice*, 58:2–6.
- Navega D, **d'Oliveira Coelho J**, Cunha E, Curate F. 2018. DXAGE: A New Method for Age at Death Estimation Based on Femoral Bone Mineral Density and Artificial Neural Networks. *Journal of Forensic Sciences*, 63(2): 497–503.

- Gonçalves D, **d'Oliveira Coelho J**, Acosta MA, Coelho C, Curate F, Ferreira MT, Gouveia M, Makhoul C, Pinto D, Oliveira-Santos I, Vassalo AR, Navega D, Cunha E. 2016. One for all and all for one: Linear regression from the mass of individual bones to assess human skeletal mass completeness. *American Journal of Physical Anthropology*, 160:427–432.
- Curate F, **d'Oliveira Coelho J**, Gonçalves D, Coelho C, Ferreira MT, Navega D, Cunha E. 2016. A method for sex estimation using the proximal femur. *Forensic Science International*, 266:579.e1–7.

Conference Papers

- Scott GR, Navega D, **d'Oliveira Coelho J**, Vlemincq-Mendieta T, Kenessey D, Pilloud MA. 2022. Tooth morphology and population affinity: testing rASUDAS2 on modern African and European-derived samples. *American Journal of Biological Anthropology*, 177(S73):164–165.
- Dorans ER, **d'Oliveira Coelho J**, Anemone RL, Bobe R, Carvalho S, Forrest F, Braun DR. 2020. Using remote sensing and machine learning to reconstruct paleoenvironmental features in the Koobi Fora Formation. *American Journal of Physical Anthropology*, 171:73–74.
- d'Oliveira Coelho J**, Anemone RL, Carvalho S. 2020. Filling Gaps in Primate Evolution: Machine Learning Approaches for Automated Fossil Site Discovery in Gorongosa, Mozambique. *Folia Primatologica*:251–252.
- Andrade B, Dias P, Santos BS, Coelho C, **d'Oliveira Coelho J**, Navega D, Ferreira MT, Wasterlain S. Morphological analysis of 3D skull models for ancestry estimation. 2018. *22nd International Conference Information Visualisation (IV)* 2018 Jul 10 (IEEE):567–573. DOI: 10.1109/iv.2018.00104.
- Makhoul C, Amarante A, **d'Oliveira Coelho J**, Navega D, Almeida M, Parker SF, Ferreira MT, Viegas DX, Marques M, de Carvalho LB, Gonçalves D, Cunha E. 2017. Experimental burning and real fire scenarios: Are they comparable in terms of chemical anthropology? *La Revue de Médecine Légale*, 8(4): 187.
- Andrade B, Dias P, Coelho C, **d'Oliveira Coelho J**, Navega D, Wasterlain SN, Ferreira MT, Santos BS. 2016. Towards automatic non-metric traits analysis on 3D models of skulls. *23rd Portuguese Meeting on Computer Graphics and Interaction (EPCGI)*:1–7. DOI: 10.1109/EPCGI.2016.7851196

Scientific Conferences

- d'Oliveira Coelho J**, Bobe R, Lourenço E, Carvalho S. 2026. A Safari in Deep Time: The Evolutionary History of the Gorongosa Ecosystem. *17th Pan-African Archaeological Association Congress (PanAF 2026): Discovering African Archaeology and Heritage Without Borders*. Maputo, Mozambique (Podium).
- Anemone RL, Bobe R, Carvalho S, Püschel T, **d'Oliveira Coelho J**. 2026. A new giant fossil hyrax from Gorongosa National Park, Mozambique. *17th Pan-African Archaeological Association Congress (PanAF 2026): Discovering African Archaeology and Heritage Without Borders*. Maputo, Mozambique (Podium).
- Boyd MD, Tagir S, Balate L, Lourenço E, Bobe R, **d'Oliveira Coelho J**, Carvalho S. 2026. The Geological Context of the Neogene fossil sites in Gorongosa National Park, Mozambique: Exploring the Southern Terminus of the East African Rift. *17th Pan-African Archaeological Association Congress (PanAF 2026): Discovering African Archaeology and Heritage Without Borders*. Maputo, Mozambique (Podium).
- Gove R, Mathe J, Bobe R, Carvalho S, Farassi R, Rabajoli E, Coelho C, **d'Oliveira Coelho J**. 2026. Dispersão de restos ósseos de animais contemporâneos em diferentes habitats do Parque Nacional da Gorongosa (PNG). *17th Pan-African Archaeological Association Congress (PanAF 2026): Discovering African Archaeology and Heritage Without Borders*. Maputo, Mozambique (Podium).
- d'Oliveira Coelho J**, Carvalho S. 2025. The case of Gorongosa National Park, a Park for the People: How primatology, science and education can be key players in nature restoration processes. *30th International Primatological Society Congress (IPS)*. Antananarivo, Madagascar (Podium).
- Carvalho S, Mathe J, Farassi RA, Reynard I, **d'Oliveira Coelho J**, Hammond P. 2025. Predation zones and primate behaviour: A study in Gorongosa National Park. *30th International Primatological Society Congress (IPS)*. Antananarivo, Madagascar (Podium).
- Anemone R, Bobe R, **d'Oliveira Coelho J**, Püschel T, Sier M, Mendes C, Archer W, Presnyakova D, Paixão E, Carvalho S. A new giant fossil hyracoid (Mammalia, Afrotheria) from the southern East African Rift in Gorongosa National Park, Mozambique. *Society for Vertebrate Paleontology Annual Conference, Journal of Vertebrate Paleontology* 81-82, Birmingham, United Kingdom, 2025 (Poster).
- Bobe R, **d'Oliveira Coelho J**, Doyle Boyd M, Lourenço E, Balate L, Tagir S, Jena P, Sanders W, Bamford M, Carvalho S. The coastal forests of eastern Africa as an evolutionary hotspot: hypotheses and new data. *Society for Vertebrate Paleontology Annual Conference, Journal of Vertebrate Paleontology* 131-132, Birmingham, United Kingdom, 2025 (Poster).

- Farassi R, **d'Oliveira Coelho J**, Carvalho S. Habitat Use and the Demographics of Object Manipulation by Wild Chacma Baboons. *30th Congress of the International Primatological Society (IPS)*. Antananarivo, Madagascar, 2025 (Poster, Awarded Best Student Poster).
- Rabajoli E, Lewis-Bevan L, **d'Oliveira Coelho J**, Coelho C, Farassi R, Mathe J, Biro D, Mubobo E, Angela M, de Salma Andicene F, Carvalho S. Rank, Relationships, and Resources: How is resource access influenced by individual ranking and social relationships in wild *Papio ursinus* in Gorongosa National Park? *30th Congress of the International Primatological Society (IPS)*. Antananarivo, Madagascar, 2025 (Poster).
- Allen G, Freymann E, d'Oliveira Coelho J, Panda A, Jaya A, Hockings KJ, Morrogh-Bernard HC. Orangutan Self-Medicative Behaviour: Feeding Ecology and Resource Combinations. *Student Conference on Conservation Science (SCCS) Cambridge*. Cambridge, United Kingdom, 2025 (Poster).
- Carvalho S, **d'Oliveira Coelho J**, Bobe R. 2024. Early hominins (and other apes) under the canopy. *14th Annual Meeting of the European Society for the Study of Human Evolution (ESHE)*. Zagreb, Croatia (Podium).
- d'Oliveira Coelho J**. Searching for Kingdon's Coastal Ape in Miocene Estuarine Forests of Gorongosa. *Joint Meeting of the International Primatological Society and the Malaysian Primatological Society*. Kuching, Malaysia, 2023 (Podium).
- Bobe R, **d'Oliveira Coelho J**, Püschel TA, Carvalho S. The diversification of African primates in the Neogene: key questions and new data. *Joint Meeting of the International Primatological Society and the Malaysian Primatological Society*. Kuching, Malaysia, 2023 (Podium).
- Farassi R, **d'Oliveira Coelho J**, Carvalho S. 2023. Object manipulation by chacma baboons (*Papio ursinus*) in Gorongosa National Park. *Joint Meeting of the International Primatological Society and the Malaysian Primatological Society*. Kuching, Malaysia, 2023 (Podium).
- d'Oliveira Coelho J**. Descoberta de sítios fósseis e reconstrução paleoecológica a partir de imagens de satélite. *Ciclo de Conferências: O Bom Antropólogo a casa torna*. Coimbra, Portugal 2023 (Seminar).
- Freymann E, **d'Oliveira Coelho J**, Muhumuza G, Hobaiter C, Huffman MA, Zuberbühler K, Carvalho S. Applying Collocation and APRIORI Analyses to Chimpanzee Diets: methods for investigating non-random food combinations in primate self-medication. *Primate Society of Great Britain Spring Meeting*. Cambridge, United Kingdom, 2023 (Podium).
- Bobe R, **d'Oliveira Coelho J**, Carvalho S. Biogeographic approaches to understanding the origin and dispersal of the earliest hominins. *Oreopithecus150*, Florence, Italy, 2022 (Podium).
- Carvalho S, Bobe R, **d'Oliveira Coelho J**, and Paleo-Primate Project. The first Miocene fossil sites of the southern end of the East African Rift: summary of key findings. *Oreopithecus150*. Florence, Italy, 2022 (Poster).
- Scott GR, Navega D, **d'Oliveira Coelho J**, Vlemincq-Mendieta T, Kenessey D, Pilloud MA. Tooth morphology and population affinity: testing rASUDAS2 on modern African and European-derived samples. *91st Annual Meeting of the American Association of Biological Anthropologists*, Denver, Colorado, USA, 2022 (Poster).
- Curate F, Navega D, Cunha E, **d'Oliveira Coelho J**. DXAGE 2.0 – An update on adult age at death estimation using bone loss. *19º Congresso Nacional de Medicina Legal e Ciências Forenses*. Coimbra, Portugal, 2021 (Poster).
- Scott GR, Navega D, **d'Oliveira Coelho J**. The Evolution of rASUDAS and Its Application to Bioarchaeological Problems. *Centro de Investigação em Antropologia e Saúde BioSocial Talks*. Online, 2021 (Seminar).
- d'Oliveira Coelho J**, Bobe R, Carvalho S. Using fossil evidence to improve our understanding of divergence estimates for the Panini and Hominini lineages. *Primate Society of Great Britain - Winter meeting*, Online, 2020 (Poster, Awarded Best Student Poster).
- Dorans ER, **d'Oliveira Coelho J**, Anemone RL, Bobe R, Carvalho S, Forrest F, Braun DR. Using remote sensing and machine learning to reconstruct paleoenvironmental features in the Koobi Fora Formation. *The 89th Annual Meeting of the American Association of Physical Anthropologists*. Los Angeles, USA, 2020 (Poster).
- d'Oliveira Coelho J**, Anemone RL, Carvalho S. Filling gaps in primate evolution: machine learning approaches for automated fossil site discovery in Gorongosa, Mozambique. *European Federation for Primatology & the Primate Society of Great Britain Winter meeting*. Oxford, United Kingdom, 2019 (Podium).
- d'Oliveira Coelho J**, Braun DR, Bobe R, Anemone RL, Carvalho S. A MaxEnt approach to estimate the likelihood of finding new fossil sites in Lake Turkana, Kenya, using satellite images. *9th Annual European Society for the study of Human Evolution meeting*. Liège, Belgium, 2019 (Poster).
- Gabriel B, Makhoul C, **d'Oliveira Coelho J**, de Carvalho LB, Cunha E. Thermally induced changes: influence of duration of heat exposure on colorimetry, morphometry and chemometrics. *FASE – Forensic Anthropology Society of Europe Advanced Course in Forensic Anthropology and One-Day Symposium 2019*. Brussels, Belgium, 2019 (Poster).
- d'Oliveira Coelho J**, Anemone RL, Bobe R, Carvalho S. Let the computers do the surveying: applying support vector machines on spectral data to identify new fossiliferous deposits in Koobi Fora, Kenya. *Paleoanthropology Society 2019 Meeting*. Albuquerque, USA, 2019 (Poster).

- d'Oliveira Coelho J**, Anemone RL, Carvalho S. Standing out in a crowd of trees: spectral outliers as predictors of fossiliferous outcrops in modern forests. *Unravelling Human Origins 2019*. York, United Kingdom, 2019 (Podium).
- d'Oliveira Coelho J**, Anemone RL, Carvalho S. Fossil sites as landscape outliers: the potential of the dbSCAN clustering algorithm for detecting fossiliferous outcrops in forests. *16th Conference of the Gesellschaft für Primatologie*. Göttingen, Germany, 2019 (Poster).
- d'Oliveira Coelho J**, Anemone RL, Carvalho S. Finding the needle in a haystack: automating spectral remote searching for hominin fossil sites in Gorongosa, Mozambique. *Proceedings of the European Society for the Study of Human Evolution*. 56. Faro, Portugal, 2018 (Poster).
- d'Oliveira Coelho J**, Anemone RL, Carvalho S. Why bother surveying? How automated search of fossil sites in Africa can help fill gaps in hominoid evolution. *Abstracts of the 27th International Primatological Society Congress*. Event-714. Nairobi, Kenya, 2018 (Podium).
- Ammer S, **d'Oliveira Coelho J**. Estimating Sex with Outline Shape Analysis of the Trochlear Constriction and the Olecranon Fossa. *American Academy of Forensic Sciences 70th Annual Scientific Meeting*. Seattle, USA, 2018 (Podium).
- d'Oliveira Coelho J**, Ammer S. Sex estimation based on outline shape analysis of the posterior distal humerus, *MORPH 2017 - A Conference on the Archaeological Applications of Morphometrics*. Aarhus, Denmark. 2017 (Podium).
- Ferreira MT, Coelho C, Rufino AI, Navega D, **d'Oliveira Coelho J**, Wasterlain SN. Os Escravos do Valle da Gafaria: recuperar a identidade através dos ossos. *Dia Internacional dos Museus*. Viseu, Portugal, 2017 (Podium).
- Rodrigues H, Fagundes L, Ramos R, Galego O, **d'Oliveira Coelho J**, Navega D, Caseiro-Alves F, Cunha E. Análise da mastóide, ouvido médio e ouvido interno em tomografia axial computadorizada (TAC) - Possível contribuição para a identificação de cadáveres. *16^o Congresso de Medicina Legal e Ciências Forenses*. Coimbra, Portugal, 2017 (Podium).
- Navega D, **d'Oliveira Coelho J**, Cunha E, Curate F. Estimativa da idade à morte com base na densidade mineral óssea do fémur proximal: uma abordagem com redes neuronais artificiais. *III Conferência do Instituto Nacional de Medicina Legal e Ciências Forenses*. Coimbra, Portugal, 2016 (Poster).
- Rufino AI, **d'Oliveira Coelho J**, Navega D. A coleção PAVd'09 (Lagos): futuro e novas perspectivas. *V Jornadas Portuguesas de Paleopatologia*. Coimbra, Portugal, 2016 (Podium).
- Coelho C, **d'Oliveira Coelho J**, Navega D, Andrade B, Dias P, Santos BS, Wasterlain SN, Ferreira MT. CraMs: uma ferramenta informática para a análise antropológica do crânio. *III Conferência do Instituto Nacional de Medicina Legal e Ciências Forenses*. Coimbra, Portugal, 2016 (Poster).
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Professional development

Used massive open online course (MOOC) platforms, to take courses offered by accredited universities and training programs in data science, programming, mathematics, natural history, among others.

Coursera, Inc. | [coursera.org](https://www.coursera.org)

Johns Hopkins University: The Data Scientist's Toolbox (100%); R Programming (100%); Getting and Cleaning Data (100%); Exploratory Data Analysis (100%); Reproducible Research (100%); Regression Models (100%); Practical Machine Learning (100%); Developing Data Products (100%); Mathematical Biostatistics Boot Camp 1 (94%); Mathematical Biostatistics Boot Camp 2 (100%); Bioinformatics: Life Sciences on Your Computer (93.5%); Statistical Analysis of fMRI Data (100%);

Stanford University: Machine Learning (100%); Statistical Learning (100%);

University of Michigan: Model Thinking (98.2%); Programming for Everybody (Python) (100%);

The University of Melbourne: Epigenetic Control of Gene Expression (94.9%);

University of North Carolina: Metadata: Organizing and Discovering Information (99.2%);

University of Alberta: Dino 101: Dinosaur Paleobiology (98.3%);

University of Minnesota: Introduction to Human Behavioral Genetics (98.6%);

Emory University: The Art and Archaeology of Ancient Nubia (99.4%);

Rice University: An Introduction to Interactive Programming in Python (98.3%);

The University of Edinburgh: Astrobiology and the Search for Extraterrestrial Life (97.8%); Introduction to Philosophy (98.2%).

edX, Inc. | [edx.org](https://www.edx.org)

Harvard University: PH525.1x Statistics and R for the Life Sciences (100%); PH525.2x Matrix Algebra and Linear Models (100%); PH525.3x Advanced Statistics for the Life Sciences (97%); PH525.4x Introduction to Bioconductor (94%); AT1x AnatomyX: Musculoskeletal Cases (98%);

MIT – Massachusetts Institute of Technology: MAS.S69x Big Data and Social Physics (100%);

The University of Queensland: World101x Anthropology of Current World Issues (98%);

Karolinska Institutet: KiexploreX Explore Statistics with R (90%);

The University of Texas at Austin: UT.7.01x Foundations of Data Analysis (93%);

University of California, Berkeley: Stat_2.3x Introduction to Statistics: Inference (88%).

FutureLearn (The Open University) | [FutureLearn.com](https://www.futurelearn.com)

University of Wollongong: *Homo floresiensis* Uncovered (100%); Bioprinting: 3D Printing Body Parts (100%)

University of Basel: Statistical Shape Modelling (90%)

University of Southampton: Archaeology of Portus (100%)

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